Technical Datasheet Input / Output Modules with Modbus RTU Protocol with RS485 Interface

The IO modules communicate via RS485. The port can drive distances up to max 700 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device).*
The RS485 Digital IO module is sturdy, low power usage and easy to use.

4 Port DO + 4 DI + 4 AI Module: -

The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. The DIP switch for Slave ID and Baud rate are placed inside the enclosure.

The design of the modules incorporates ‘**resettable Fuses**’ to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	12 Pin 5.08 mm pitch pluggable screw Terminal Block, 2 Pin 5.08 mm pitch pluggable screw terminals.
Dimensions	110 mm L x 110 mm B x 50 mm H
Power	Input Power – 12 - 24 VDC or 24 V AC/DC Typical – 12V DC @ 120mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Certifications	RoHS, APPROVED CE APPROVED

DO with Relay Output –

Channels	4
Contact Form	1A, 1C
Contact Material	Ag Alloy
Contact Capacity	10A @ 240VAC, 10A @ 28VDC
Coil Voltage	5 – 48 VDC
Coil Power	0.36 W
Insulation Resistance	250MΩ
Electrical Life	1 x 10^5
Mechanical Life	1 x 10^7
Operating Time	7 msec, Max 15 msec
Release Time	2 msec, Max 6 msec

DI Inputs –

Channels	4
Sense Voltage	3.3 – 30 VDC
Sense Logic	Logic ‘0’ = <1 VDC, Logic ‘1’ = >3.3 VDC
Isolation	Optically isolated
Response Time	2 msec, Max 6 msec

AI Inputs –

Channels	4
Input Signal	4 – 20 mA / 0 – 10VDC / NTC Type 3 (jumper selectable)
Accuracy	± 2% Full scale
Input Resolution	10 bits / 12 bits (Optional)
Isolation	Optically Isolated
External Loop Voltage	+ 12 VDC min
AI input impedance	120Ω

Additional Features: -

All inputs and communication ports isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
750V isolation.
CRC Error check.
No configuration needed on the IO board

Configuration Settings: -

Communication Speed	9600 – 115200 (DIP SW selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Configurable with DIP Switch
Function code DO	0x05 and 0x0F (5 Single coil & 15 multiple coil)
DO Register Address	0,1,2,3
Function code DI	0x02 Read discrete Input
DI Register Address	4,5,6,7
Function Code AI	0x03 Read Holding Registers
AI Register Address	10 Bit - 2,3,4,5 / 12 Bit – 6,7,8,9.

ID	Function Description	Register Description	Modbus Function Code	Protocol	Data Type
1	**DO 1**	00001	0X05,0X0F	RS485	1 Bit Boolean
2	**DO 2**	00002	0X05,0X0F	RS485	1 Bit Boolean
3	**DO 3**	00003	0X05,0X0F	RS485	1 Bit Boolean
4	**DO 4**	00004	0X05,0X0F	RS485	1 Bit Boolean
5	**DI 1(With & Without Potential)**	10005	0X02	RS485	1 Bit Boolean
6	**DI 2(With & Without Potential)**	10006	0X02	RS485	1 Bit Boolean
7	**DI 3(With & Without Potential)**	10007	0X02	RS485	1 Bit Boolean
8	**DI 4(With & Without Potential)**	10008	0X02	RS485	1 Bit Boolean
9	**AI 1 – 10 Bit**	40002	0X03	RS485	16 Bit Unsigned int
10	**AI 2 – 10 Bit**	40003	0X03	RS485	16 Bit Unsigned int
11	**AI 3 – 10 Bit**	40004	0X03	RS485	16 Bit Unsigned int
12	**AI 4 – 10 Bit**	40005	0X03	RS485	16 Bit Unsigned int
13	**AI – 1 12 Bit**	40006	0x03	RS485	16 Bit Unsigned int
14	**AI – 2 12 Bit**	40007	0x03	RS485	16 Bit Unsigned int
15	**AI – 3 12 Bit**	40008	0x03	RS485	16 Bit Unsigned int
16	**AI – 4 12 Bit**	40009	0x03	RS485	16 Bit Unsigned int

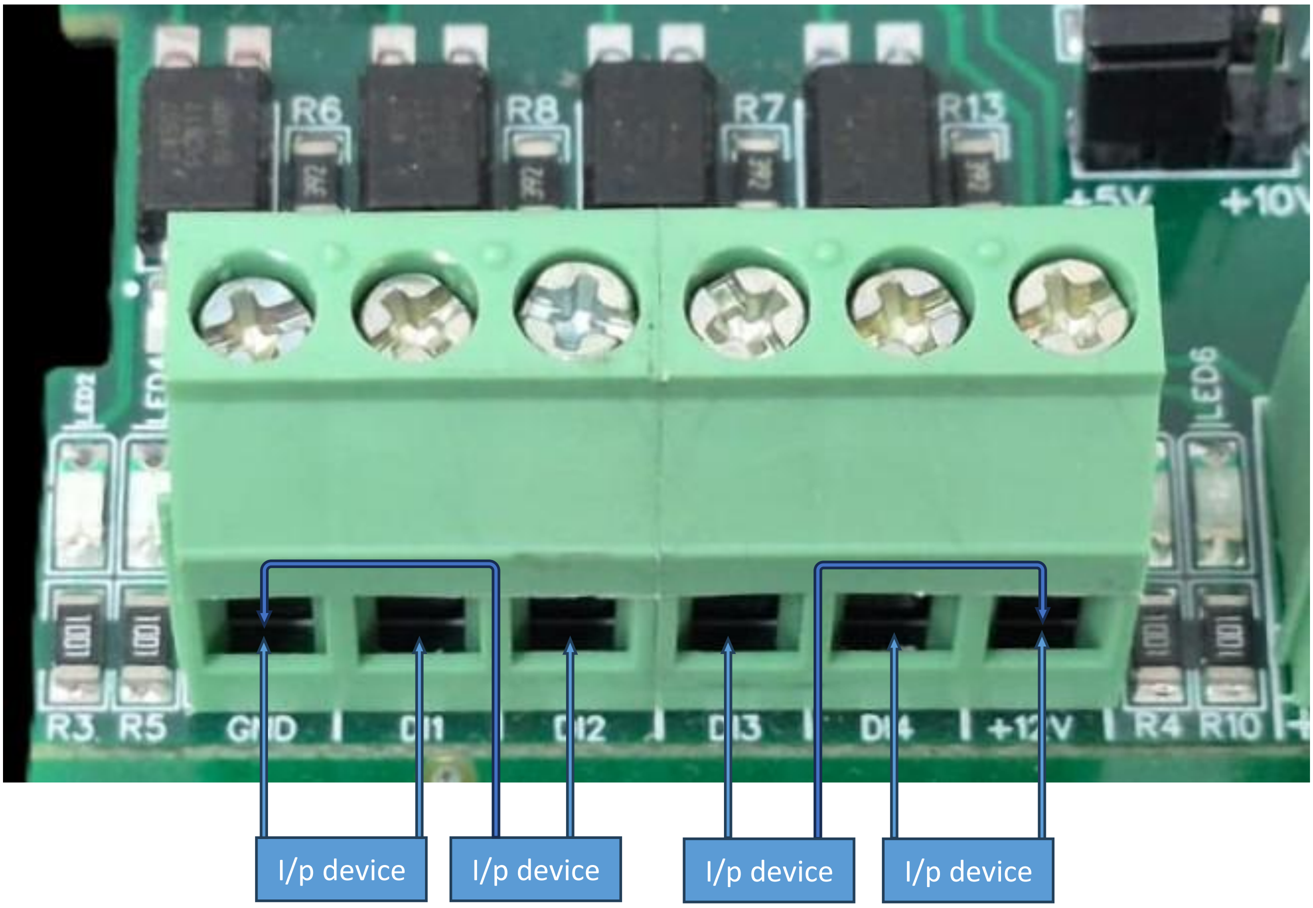
Note: -

For MODBUS communications, **a shielded and twisted pair cable** is used. One example of such cable is Belden 3105A.

Recommended Cable Electrical Characteristics: -

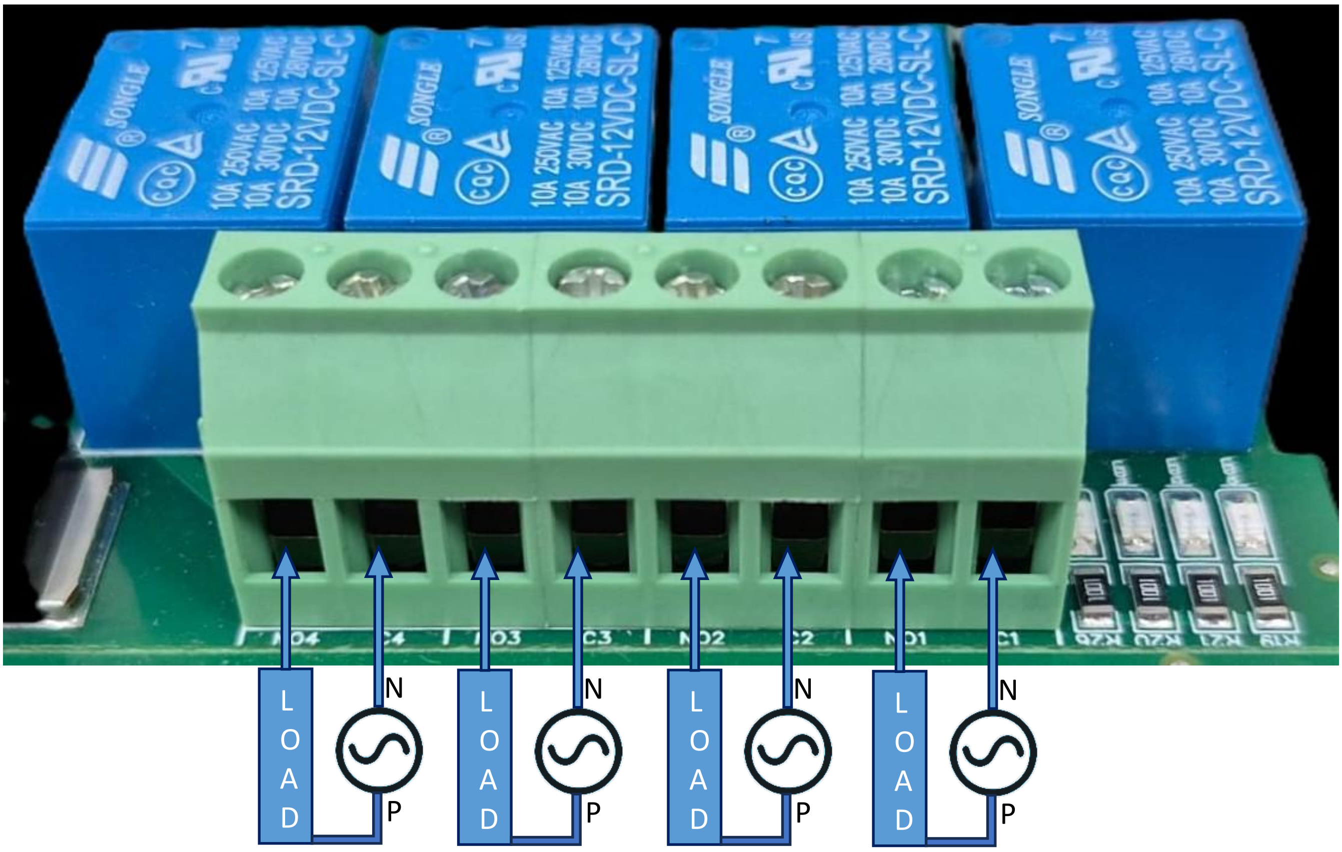
22 AWG Cable	Shielded and twisted pair should be used.
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100ft

DI



- **Operation:** The module provides four isolated input channels (**DI1–DI4**) for sensing external signals or dry contacts. These inputs are opto-isolated for circuit protection.
- **Terminal Layout:**
  - **GND:** Common Ground reference.
  - **DI1 – DI4:** Individual Digital Input channels.
  - **+12V:** Internal reference voltage for sourcing "Dry Contact" switches.
- **Wiring Options:**
  - **Dry Contact:** Connect a switch between the **+12V** terminal and any **DI** pin.
  - **External Voltage:** Connect the external signal to a **DI** pin and the external ground to the **GND** terminal.
- **Voltage Rating:** The input circuitry is rated for a maximum of **80VDC**. Ensure external signals do not exceed this limit to prevent damage to the isolation ICs.

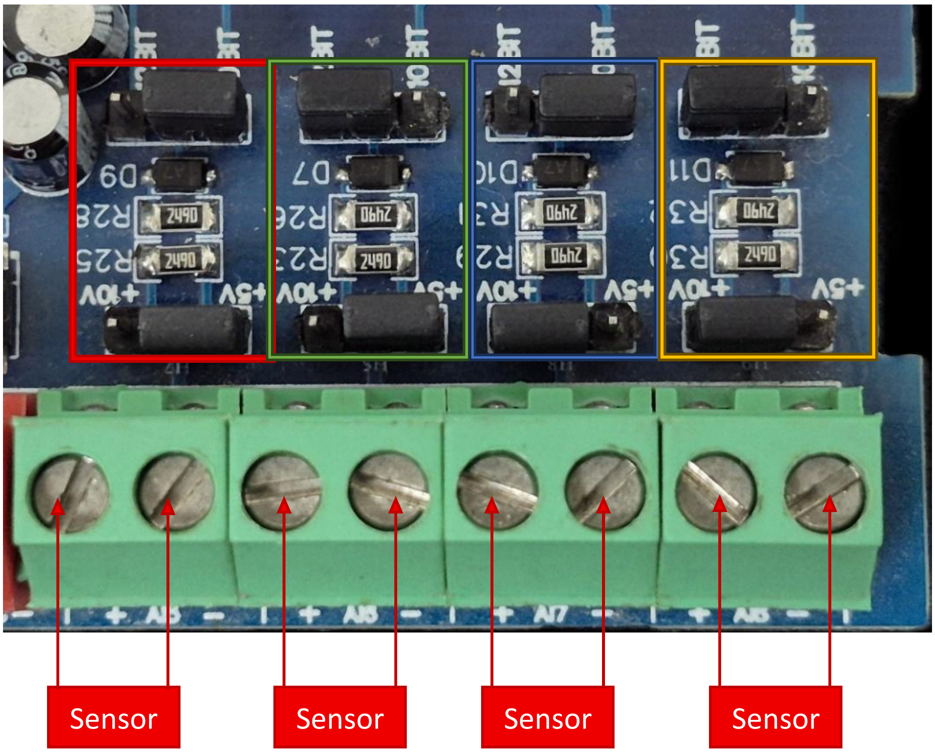
DO



- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C is internally connected to NO**, allowing current to flow. When the relay is de-energized, the connection between **C and NO remains open**, and the load stays OFF.
- **Load Wiring Connections:**
    - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The **load is connected between terminal NO and Neutral (N)**.
    - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The **load is connected between terminal NO and Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

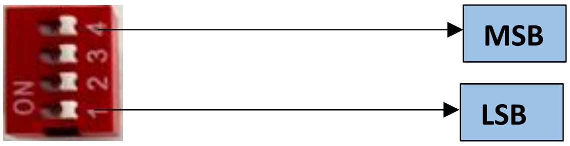
  Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or 10A @ 125VAC) or **10A @ 30VDC** (or 10A @ 28VDC).

AI



- **Analog Input (AI) Operation:** Each channel provides two terminals: **+** (Positive) and **-** (Negative/Common). These are used to read external analog sensor signals.
- **Channel Configuration (Visual Guide):** Each of the four AI channels features two dedicated jumpers to independently set the voltage range and bit resolution. Refer to the color-coded boxes in the board diagram to configure your desired scenario:
    - **Red Box (5V, 10-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **10 BIT**.
    - **Green Box (5V, 12-bit):** Set the voltage jumper to **+5V** and the resolution jumper to **12 BIT**.
    - **Blue Box (10V, 10-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **10 BIT**.
    - **Yellow Box (10V, 12-bit):** Set the voltage jumper to **+10V** and the resolution jumper to **12 BIT**.
- **Sensor Wiring Connections:**
    - The analog sensor's positive signal wire should be connected to the **+** terminal of the desired channel.
    - The sensor's common or ground wire should be connected to the corresponding **-** terminal.
- **Crucial Setup Recommendation:** Always ensure that the voltage jumper setting strictly matches the maximum output range of your connected sensor (either 5V or 10V) to guarantee accurate analog-to-digital conversion and prevent data clipping.

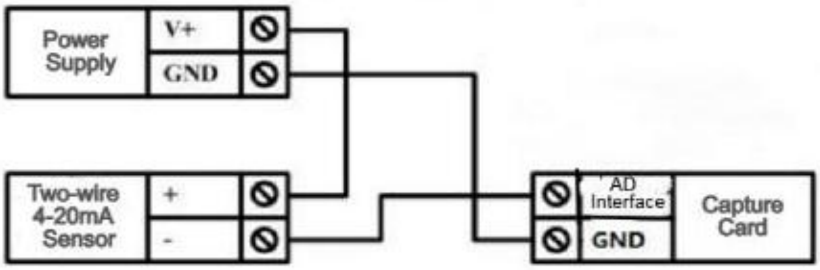
SLAVE ID DESCRIPTION



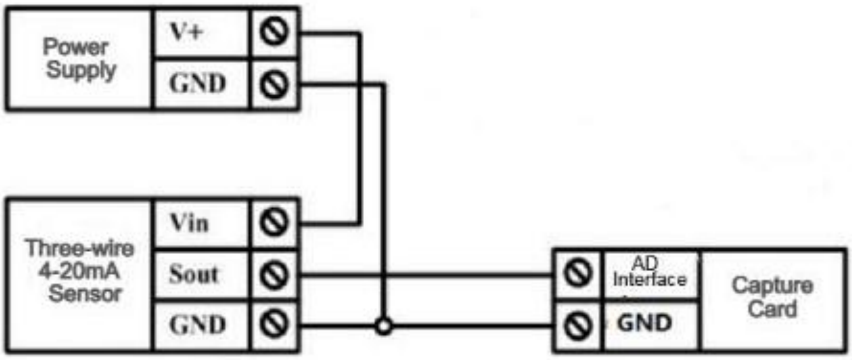
For Slave ID Selection SW is used to Set The SLAVE ID .
For Slave ID DIP Switch LSB is "1" follow through "4" is MSB.
Slave ID Confirmed through below Device ID table .
IF Eg. Slave ID 1 is Needed to be selected Switch number 1 should pulled up other three should be selected down side. So"1 0 0 0" will be selected as Slave ID 1.

Slave ID	DIP SWITCH				OUTPUT (Binary)	OUTPUT (Decimal)
	1	2	3	4		
0	OFF(0)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
1	ON(1)	OFF(0)	OFF(0)	OFF(0)	0 0 0 1	1
2	OFF(0)	ON(1)	OFF(0)	OFF(0)	0 0 1 0	2
3	ON(1)	ON(1)	OFF(0)	OFF(0)	0 0 1 1	3
4	OFF(0)	OFF(0)	ON(1)	OFF(0)	0 1 0 0	4
5	ON(1)	OFF(0)	ON(1)	OFF(0)	0 1 0 1	5
6	OFF(0)	ON(1)	ON(1)	OFF(0)	0 1 1 0	6
7	ON(1)	ON(1)	ON(1)	OFF(0)	0 1 1 1	7
8	OFF(0)	OFF(0)	OFF(0)	ON(1)	1 0 0 0	8
9	ON(1)	OFF(0)	OFF(0)	ON(1)	1 0 0 1	9
10	OFF(0)	ON(1)	OFF(0)	ON(1)	1 0 1 0	10
11	ON(1)	ON(1)	OFF(0)	ON(1)	1 0 1 1	11
12	OFF(0)	OFF(0)	ON(1)	ON(1)	1 1 0 0	12
13	ON(1)	OFF(0)	ON(1)	ON(1)	1 1 0 1	13
14	OFF(0)	ON(1)	ON(1)	ON(1)	1 1 1 0	14
15	ON(1)	ON(1)	ON(1)	ON(1)	1 1 1 1	15

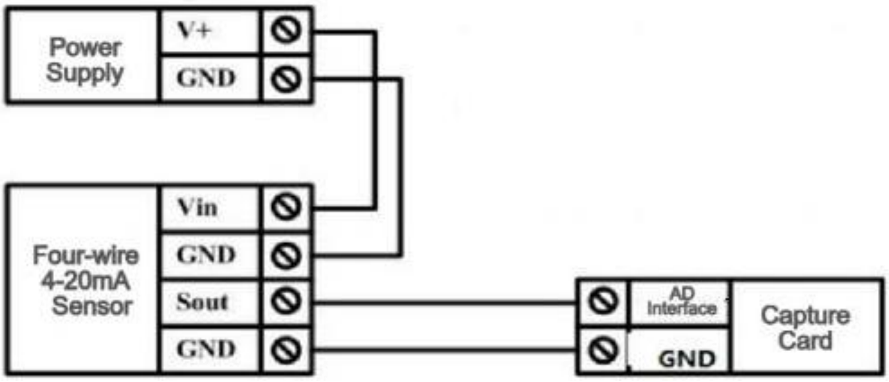
Two-wire Sensor Wiring Diagram

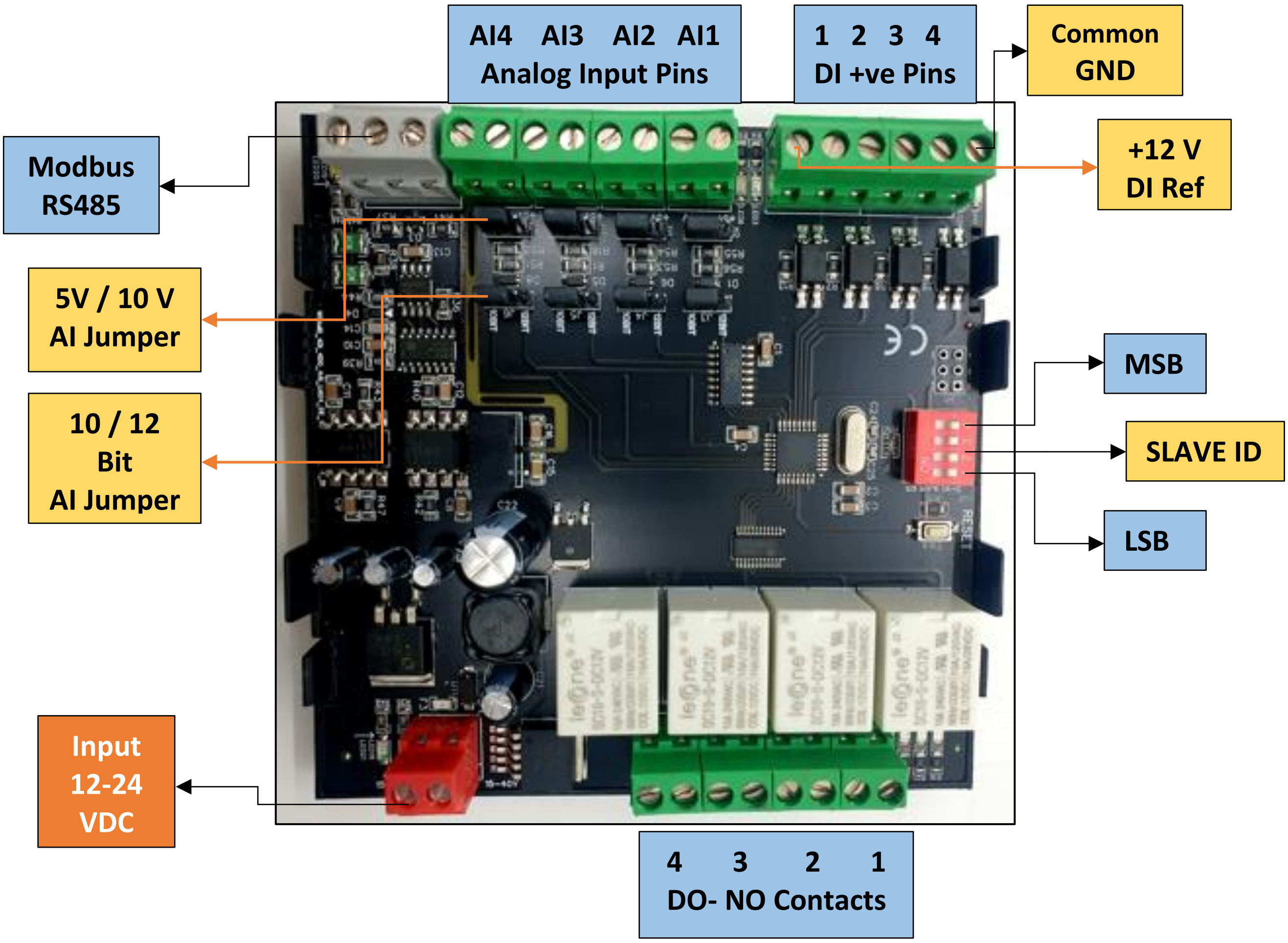


Three-wire Sensor Wiring Diagram



Four-wire 4-20mA Sensor





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